

Preserving and improving the legacy of eScience: the GLOBO experience

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Abstract—Legacy scientific codes are essential to long-standing operational workflows but often cannot fully exploit modern heterogeneous high-performance computing (HPC) architectures. GLOBO, a global atmospheric circulation model developed at CNR-ISAC, is one such model that is still used for research and operational weather forecasting. In this work, we present an ongoing refactoring of GLOBO to improve its scalability and prepare it for GPU acceleration and cloud execution. The modernization focuses on three main areas: dynamic memory allocation, replacement of legacy point-to-point MPI communications with optimized collective operations, and OpenACC-based GPU offloading. We describe the redesign of the communication strategy, which replaces multiple `MPI_Isend/MPI_Irecv` exchanges with collective primitives (`MPI_Bcast`, `MPI_Scatterv`, `MPI_Sendrecv`, `MPI_Reduce`) to reduce overhead and improve parallel efficiency. Scalability experiments across multiple spatial resolutions show that a new collective communication strategy consistently matches or exceeds the original implementation, delivering up to 40% runtime reduction in high-resolution, multi-node configurations. These results lay the groundwork for refactoring GLOBO toward a fully dynamic, GPU-accelerated, cloud-ready execution, preserving its scientific heritage while ensuring sustainable performance on future HPC platforms.

Index Terms—Legacy codes, MPI collectives, HPC, Scalability, GPU offloading, Weather modeling

I. INTRODUCTION

Modernizing legacy scientific codes is a central challenge in the era of heterogeneous high-performance computing (HPC). Many weather and climate models, originally developed for CPU-only architectures, lack the modularity and parallel scalability required to leverage today's multi-GPU systems.

The GLOWPP project, aims to refactor and optimize a global weather prediction model for sustainable use on both HPC clusters and cloud platforms. The goal is to bridge the gap between traditional science and modern computing by combining reproducibility, portability, and performance.

II. THE GLOBO MODEL

GLOBO [1] is a global, hydrostatic, atmospheric general circulation model developed at the Institute of Atmospheric Sciences and Climate of the National Research Council of Italy (CNR-ISAC). It is the global version of the Bologna Limited Area Model (BOLAM) [2]. GLOBO is currently used both as a research tool for predictability studies and as an operational forecasting tool¹. It is part of the CNR-ISAC modeling chain to produce daily 7-day global forecasts and is used weekly to run a 40-member ensemble to produce global sub-seasonal probabilistic forecasts [3], contributing to the sub-seasonal to seasonal (S2S) initiative. The model is based on a hydrostatic dynamical core on a regular, latitude/longitude staggered Arakawa C-grid in physical space, using explicit time-stepping. Its flexible design allows the model to work with a wide range of horizontal resolutions, from 312 km down to 19 km, while the vertical resolution can be adjusted from 50 to 70 levels.

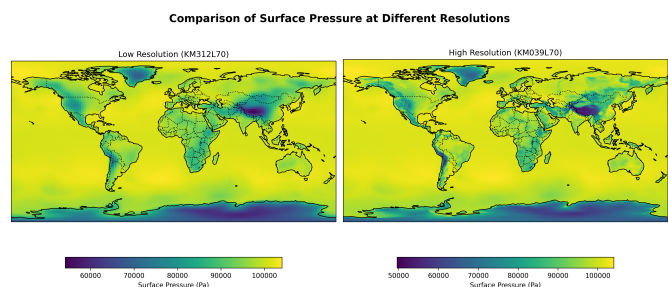


Fig. 1. Surface pressure fields simulated by GLOBO at different resolutions. *Left*: Low resolution (KM312L70). *Right*: High resolution (KM039L70).

III. DESIGN AND IMPLEMENTATION STRATEGY

The restructuring of the GLOBO model is organised in three main lines of development: (i) the introduction of dynamic

¹<https://www.isac.cnr.it/dinamica/projects/forecasts/>

memory allocation to reduce static resource usage and improve flexibility in domain decomposition, (ii) the replacement of point-to-point MPI communication with optimised collective operations to increase scalability on multi-node architectures, and (iii) the preparation of code for GPU acceleration via OpenACC directives.

Currently, the work mainly focused on the second point, with a major overhaul of the communication layer. Originally, the distribution of global parameters and subdomain data was done through iterative loops in which only the root pprocess sent messages to all other processes using multiple `MPI_Isend/MPI_Irecv` calls, followed by `MPI_Wait`. This sequential scheme generated bottlenecks and high synchronization costs, particularly for multi-node, high-resolution configurations. Instead, the new implementation adopts a strategy based on collective communications: `MPI_Bcast` is used for global parameter dissemination, `MPI_Scatterv` for the distribution of sub-domains of varying sizes in a single operation, and `MPI_Sendrecv` for simultaneous boundary exchanges. Reduce operations are now handled collectively via `MPI_Reduce`, while the elimination of non-blocking calls followed by `MPI_Wait` has simplified the code and improved maintainability. These changes represent the first step in GLOBO's modernization, establishing the foundation for dynamic memory allocation and GPU offloading.

Globo Model Communication

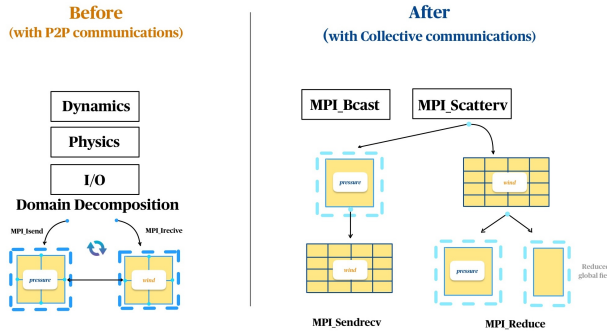


Fig. 2. Refactoring of GLOBO's communication strategy. *Left*: Original point-to-point MPI exchanges (`MPI_Isend`, `MPI_Irecv`) for pressure and wind data. *Right*: Refactored collective operations (`MPI_Bcast`, `MPI_Scatterv`, `MPI_Sendrecv`, `MPI_Reduce`) improving scalability.

IV. EARLY RESULTS AND EVALUATION

The impact of replacing point-to-point communication with MPI collectives was evaluated through highly scalable experiments on four spatial resolutions. Figure 3 summarizes the best execution times for both communication strategies.

At coarse resolutions (KM312L70 and KM156L70), both approaches show nearly identical performance, indicating that communication overhead is negligible compared to computation. At intermediate resolution (KM078L70), collective communication achieves stable execution times up to 192 processes, while point-to-point communication shows slightly

higher variability and slower performance with a large number of processes. At the highest resolution (KM039L70), collective communications offer a significant improvement, reducing execution time from 6,736 to 3,939 s (a 40% speed increase), highlighting their advantage in terms of scalability in large-scale fine-grained executions. In general, collective MPI operations improve communication efficiency in GLOBO when scaling to high-resolution domains and a large number of processes, mitigating the overhead and synchronization costs of traditional point-to-point exchanges.

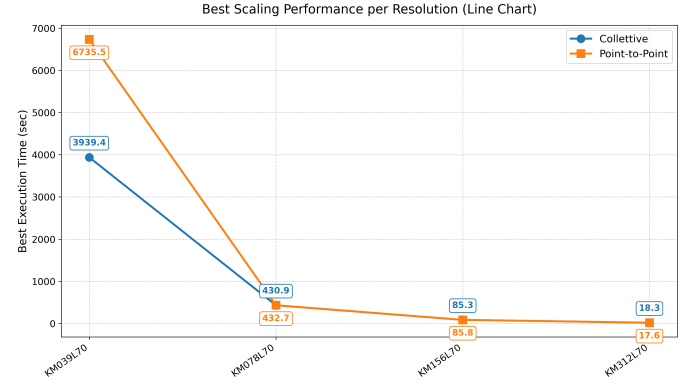


Fig. 3. Best execution time across different resolutions comparing collective and point-to-point communication strategies.

V. CONCLUSION

The integration of collective MPI communications into GLOBO simplifies code management and improves scalability on distributed memory HPC systems. Experimental results show that collective communications consistently match or exceed point-to-point communications, with significant gains in large-scale, high-resolution simulations. Although we have presented the redesign of domain decomposition-related communications here, the optimization of ghost boundaries collective communication is ongoing. This refactoring lays a solid foundation for future optimization steps, including dynamic memory management and GPU offloading, paving the way for production-scale execution of GLOBO on heterogeneous HPC and cloud platforms.

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